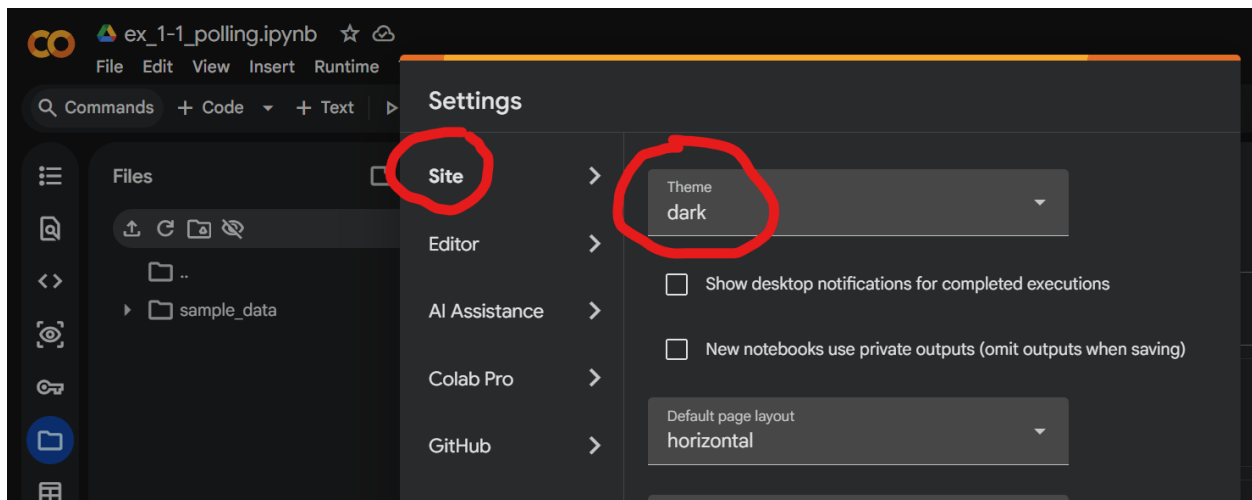


Exercise 1-1 to Google Colab.

Getting Started & Setting Themes

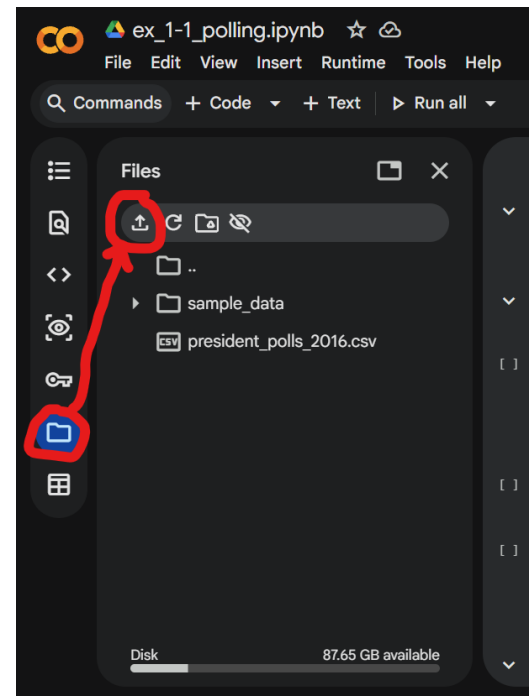
1. You will need two files to complete this lab. Download these files from Github or Canvas.
Chapter01/Exercise 1.1 Polling.ipynb
Chapter01/president_polls_2016.csv
2. **Open Google Colab:** Navigate to colab.research.google.com. Login with either your @maricopa.edu account, or your personal account. I suggest using your personal account, so that you retain all your class files after you finish your studies.
3. **Set the Theme:** Navigate to **Tools** → **Settings** → **Site** tab. Under **Theme**, toggle between **Light** and **Dark**. Colab Light is recommended to match the textbook screenshots. The instructor prefers Colab Dark. The choice is yours!



Opening Files & Environment Setup

4. Load the Notebook and Data:

- Click **File** → **Upload notebook** to upload the Chapter 01/Exercise 1.1 Polling.ipynb file.
- Click the **Folder icon** on the left sidebar (File Browser). Upload the file from Chapter 01/president_polls_2016.csv file to the runtime. You may first need to click on "Connect" at the top right of the notebook to connect to the compute runtime.
- After the file is uploaded, a warning will appear. Files that are uploaded to the runtime may be deleted at any time. If this happens while you're working on the notebook, you'll have to upload the president_polls_2016.csv file again.



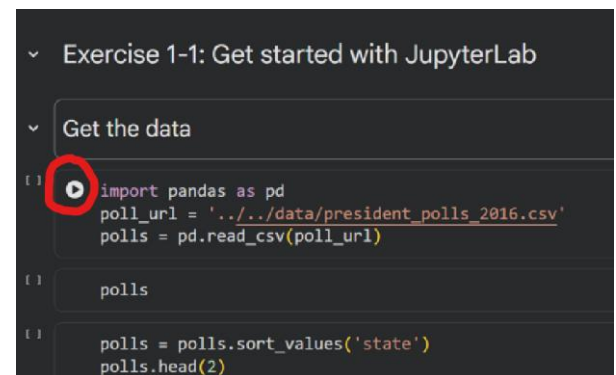
5. Clear Outputs: Go to **Runtime** → **Disconnect and delete runtime** (or **Restart session**) to clear state and outputs. If these options are greyed out, you may skip this step.

6. Run Cell 1: Click the **Play button** on the left side of the first code cell (or press Ctrl+Enter). Ensure the path in poll_url matches where you uploaded your CSV file. You will need to change this to:

```
poll_url = 'president_polls_2016.csv'
```

7. Run Cell 2: Click inside the cell that contains the command "polls". Press Shift+Enter to run the cell to view the polls DataFrame. Modify the code to `polls.head()` and press Shift+Enter again to view only the first 5 rows.

8. Run Cell 3: Run the cell containing `polls.sort_values('state')`.



Debugging Syntax & Runtime Errors

9. In the third cell, intentionally break the syntax by editing the code to `polls.sort_values('stat` (removing the `e` and closing parenthesis `)`).

10. Press Shift+Enter.
Colab will display a red **SyntaxError** trace at the bottom of the cell.

11. Add the missing closing

parenthesis `)` and run it again. You will now see a **KeyError: 'stat'** (runtime error). Correct `'stat'` back to `'state'` and run the cell to execute the sort correctly.

```
[5]
0s
polls = polls.sort_values('stat
polls.head(2)

File "/tmp/ipykernel_3022/1278912162.py", line 1
    polls = polls.sort_values('stat
                              ^
SyntaxError: unterminated string literal (detected at line 1)
```

```
[6]
0s
polls = polls.sort_values('stat')
polls.head(2)

-----
KeyError                                Traceback (most recent call last)
/tmp/ipykernel_3022/1368314204.py in <cell line: 0>()
----> 1 polls = polls.sort_values('stat')
      2 polls.head(2)

-----
1 frames
/usr/local/lib/python3.13/dist-packages/pandas/core/generic.py in _get_label_or_level_val
axis)
   1909         values = self.axes[axis].get_level_values(key)._values
   1910     else:
-> 1911         raise KeyError(key)
   1912
   1913     # Check for duplicates

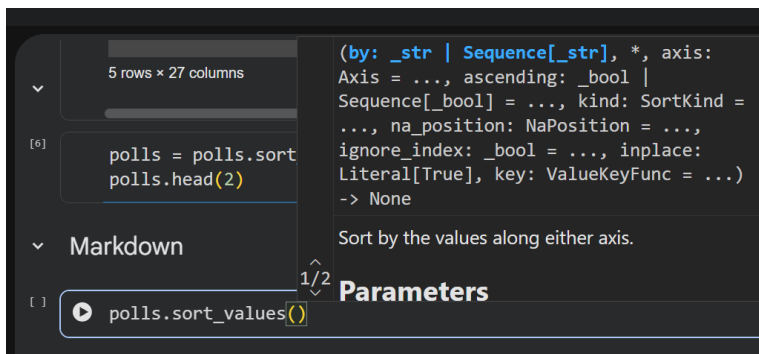
KeyError: 'stat'
```

Markdown, Tab Completion, and Code Intelligence

12. **Add Markdown Subheading:** Click inside the `sort_values()` cell that you were just working on. Click **+ Text** (or press Ctrl+M Ctrl+B then Ctrl+M Ctrl+M).

Use Markdown language, using textbook figure 1-11 as a guide, to create a subheading that says “Use Tab completion and tooltips”.

13. **Tab Completion:** Add a **+ Code** cell below the text cell. Type `polls.s` and wait briefly for Colab’s automatic pop-up recommendation. Type the “o” to refine the list further. Select `sort_values` from the drop-down list by clicking it directly, or by using the arrow keys to select it and pressing the tab key. Do not hit enter yet!
14. **Tooltips / Documentation:** Now type a left parenthesis. This will trigger a pop-up card showing the function signature and parameter documentation.



15. **Method Chaining:** Complete the line to read:
`polls.sort_values('state', ascending=False)`
and run the cell.

Now go back and “chain” (or add) `.head()` at the end.

The following should now be the code in the entire cell:

```
polls.sort_values('state', ascending=False).head()
```

and run the cell.

16. **Inspect Chained Methods:** Hover over `sort_values` to see its documentation. Notice that hovering over chained methods at the end may require placing the cursor directly on that specific function name.

Running the Rest of the Notebook

17. **Run Remaining Cells:** Select **Runtime** → **Run cell and below** or press Shift+Enter cell by cell. Plots created with matplotlib or pandas will render inline directly below the code cells.
18. **Review Code:** Review the code in each of the cells that you've just run. Note the use of a dictionary in the rename() method and the use of lists in some of the other statements. Note too that the plot() method in the last cell of this Notebook is chained to the query() method.

Submit your Exercise for grading

19. Add a text/markdown cell at the top of your notebook that contains the following:
Your Name (Replace with your actual name)
Exercise 1-1
Class (AIM111 – Introduction to Data Science)
Date (Today's Date)
20. Select **File** → **Download** → **Download .ipynb**
21. Submit the file in Canvas for this assignment.